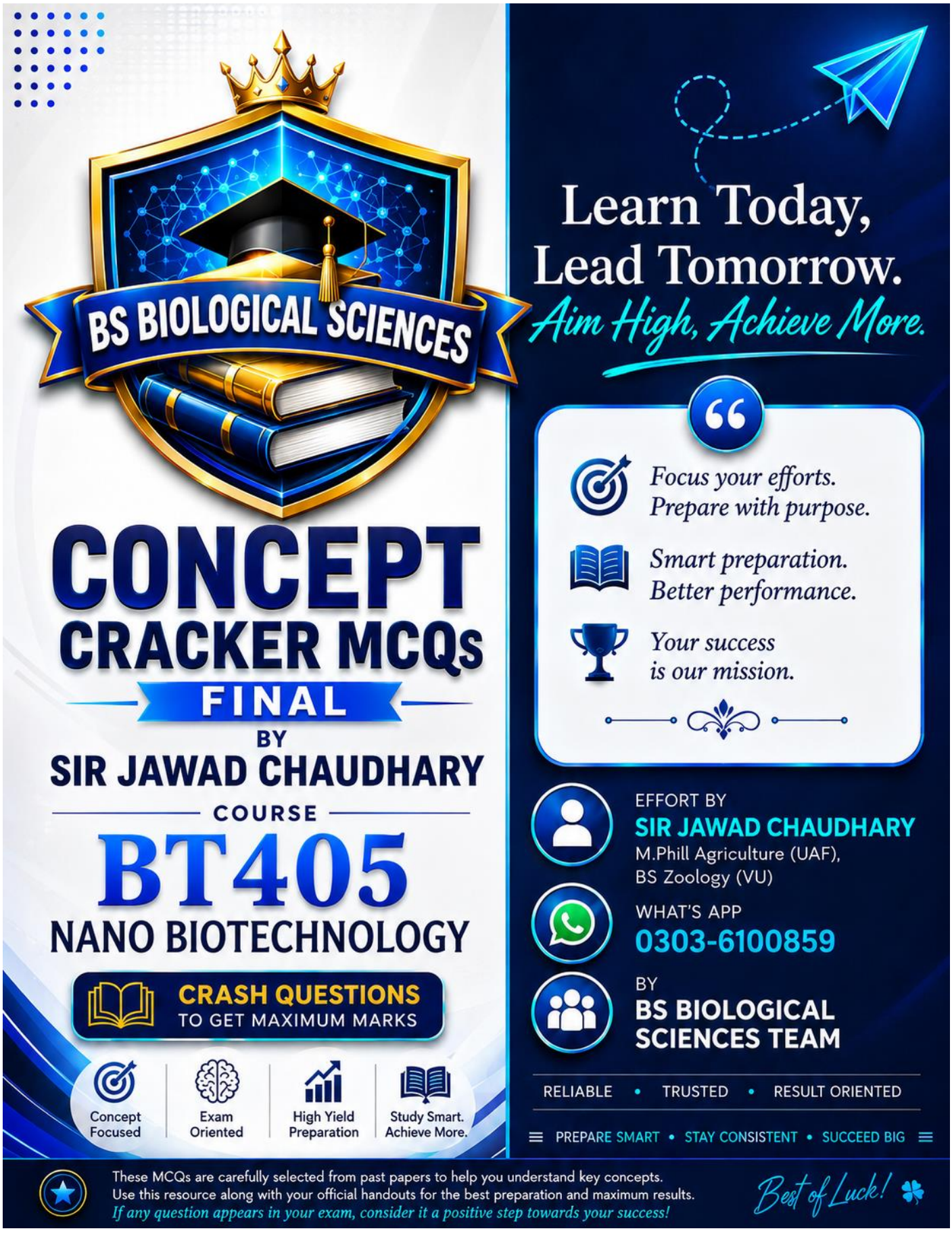


JAWAD CHAUDHARY

MOST FREQUENTLY REPEATED CONCEPTUAL AREAS IN BT405 NANO BIOTECHNOLOGY CONCEPT CRACKER MCQS FINAL EXAMS

Main Group (BS Biological Sciences) https://chat.whatsapp.com/DWXwJpXahbxFVQutey8JXN

BT405 Nano Biotechnology https://chat.whatsapp.com/GShFsFcEnlCB4hfz4GB9HK

Q1: The most common working definition of nanoscience is the study of phenomena and manipulation of materials at atomic, molecular and macromolecular scales.

Answer: Nanoscience

Q2: Nanotechnologies are the design, characterization, production and application of structures, devices and systems by controlling shape and size at the _____ scale.

Answer: Nanometre

Q3: The nanoscale is considered to cover the range from _____ to _____ nm.

Answer: 1–100

Q4: One nanometer (nm) is equal to one _____ of a meter.

Answer: Billionth

Q5: A nanomaterial is an object that has at least one dimension in the nanometre scale, approximately _____ to _____ nm.

Answer: 1–100

Q6: When particles assume nanoscale dimensions, their properties can differ significantly from those at the _____ scale.

Answer: Larger

Q7: At nanoscale dimensions, the principles of classic physics are no longer enough and _____ mechanics becomes important.

Answer: Quantum

Q8: Nanotechnology promises breakthroughs in materials and manufacturing, nanoelectronics, medicine and healthcare, energy, biotechnology, information technology, and _____ security.

Answer: National

Q9: Nanomaterials produced deliberately through a defined fabrication process are called _____ nanomaterials.

Answer: Intentionally-made

Q10: Nano-sized particles that occur naturally in the environment or are produced without intention are called _____ nanomaterials.

Answer: Non-intentionally-made

Q11: A concept system is often called an _____.

Answer: Ontology

Q12: A set of essential characteristics that come together as a unit to form a concept is called _____.

Answer: Intension

Q13: A set of objects abstracted into a concept is called the _____.

Answer: Extension

Q14: Most terms in nanotechnology are normally prefixed by _____.

Answer: Nano

Q15: One of the nanotechnology language terms mentioned in the handout is _____.

Answer: Nanobots

Q16: Nanotechnology makes products lighter, stronger, faster, smaller and more _____.

Answer: Durable

Q17: Devices at the nanoscale need less material, use less energy and may have an extended range of _____.

Answer: Accessibility

Q18: Lab-on-a-chip technology enables real-time point-of-care _____.

Answer: Testing

Q19: Nanomaterial surfaces on implants improve wear and resist _____.

Answer: Infection

Q20: Nanoparticles in pharmaceutical products improve absorption and make drugs easier to _____.

Answer: Deliver

Q21: Carbon nanotubes help sports equipment become stronger and _____.

Answer: Lighter

Q22: Water filters that are only 15–20 nanometers wide can remove virtually all _____ and bacteria.

Answer: Viruses

Q23: Most sunscreens today are made from nanoparticles that effectively absorb the more dangerous _____ range.

Answer: Ultraviolet

Q24: Plastics containing nanoclays increase resistance to oxygen, carbon dioxide and _____ permeation.

Answer: Moisture

Q25: Nanotechnology allows chemical sensors to detect a single molecule out of _____.

Answer: Billions

Q26: The concept of nanotechnology is attributed to Richard Feynman's vision of working atom by _____.

Answer: Atom

Q27: Professor Taniguchi used the word nanotechnology to describe processing or building parts with nanometric _____.

Answer: Tolerances

Q28: Carbon 60 was named after Richard Buckminster Fuller and is also called _____.

Answer: Buckyballs

Q29: The top-down method in nanotechnology is used by computer chip _____.

Answer: Manufacturers

Q30: The bottom-up approach to manufacturing is analogous to the way biological systems are _____.

Answer: Made

Q31: Biotechnology is the use of living organisms, or the products of living organisms, for human _____.

Answer: Benefit

Q32: Biotechnology is multidisciplinary and involves engineering, computer science, cell and molecular biology, microbiology, genetics, physiology, biochemistry, immunology and _____.

Answer: Virology

Q33: Recombinant DNA technology involves genetic manipulation of bacteria, viruses, fungi, plants and _____.

Answer: Animals

Q34: Ancient biotechnology is related to food and shelter, including _____.

Answer: Domestication

Q35: Classical biotechnology is built on ancient biotechnology and fermentation promoted food production and _____.

Answer: Medicine

Q36: Modern biotechnology manipulates genetic information in organisms and is called genetic _____.

Answer: Engineering

Q37: One example of modern biotechnology given in the handout is the development of disease-resistant _____.

Answer: Plants

Q38: Golden rice was engineered to be more _____.

Answer: Nutritious

Q39: Genetically engineered bacteria can degrade environmental _____.

Answer: Pollutants

Q40: Microbial biotechnology involves manipulation of microorganisms such as yeast and _____.

Answer: Bacteria

Q41: Microbial biotechnology can be used to create better _____.

Answer: Enzymes

Q42: Microbial biotechnology also helps in more efficient decontamination of industrial waste _____.

Answer: Product removal

Q43: Microbes are used to clone and produce batch amounts of important _____.

Answer: Proteins

Q44: The United Nations Food and Agricultural Organization predicts that by 2050 the world population will reach _____ billion.

Answer: 9.1

Q45: In animal biotechnology, animals can be a source of medically valuable proteins, antibodies and _____ animals.

Answer: Transgenic

Q46: Female transgenic animals can express therapeutic proteins in their _____.

Answer: Milk

Q47: Forensic biotechnology includes DNA fingerprinting for paternity cases and identification of human _____.

Answer: Remains

Q48: Bioremediation is the use of biotechnology to process and degrade substances that contribute to environmental _____.

Answer: Pollution

Q49: Aquaculture is the raising of finfish or shellfish in controlled conditions for use as food _____.

Answer: Sources

Q50: Genentech developed _____ in 1982, the first biotech drug to be FDA approved.

Answer: Humulin

Q51: Modern biotechnology became possible after the discovery of recombinant _____ technology.

Answer: DNA

Q52: Recombinant DNA technology enables scientists to modify the genetic material of _____.

Answer: Organisms

Q53: One major objective of agricultural biotechnology is to improve crop _____.

Answer: Productivity

Q54: Disease-resistant plants are produced mainly through genetic _____.

Answer: Engineering

Q55: Increasing the nutritional value of crops is an application of _____ biotechnology.

Answer: Agricultural

Q56: Golden rice was developed to address deficiencies of specific _____.

Answer: Nutrients

Q57: The use of microorganisms to manufacture valuable products is called _____ biotechnology.

Answer: Microbial

Q58: Improved industrial enzymes are commonly produced using genetically modified _____.

Answer: Microorganisms

Q59: Biotechnology can improve industrial processes by increasing product _____.

Answer: Efficiency

Q60: Large-scale production of recombinant proteins commonly utilizes _____.

Answer: Microbes

Q61: Animal biotechnology contributes to the production of therapeutic _____.

Answer: Proteins

Q62: Transgenic animals are produced by introducing foreign _____.

Answer: Genes

Q63: Expression of pharmaceutical proteins in milk is known as molecular _____.

Answer: Farming

Q64: DNA fingerprinting is based on variations present within an individual's _____.

Answer: DNA

Q65: One important application of forensic biotechnology is criminal _____.

Answer: Investigation

Q66: Bioremediation primarily uses living organisms to remove environmental _____.

Answer: Pollutants

Q67: Aquaculture provides an alternative source of aquatic _____.

Answer: Food

Q68: One advantage of aquaculture is increasing the availability of seafood for human _____.

Answer: Consumption

Q69: Biotechnology integrates biological sciences with _____ sciences.

Answer: Engineering

Q70: Manipulation of microorganisms for industrial use is an example of _____ biotechnology.

Answer: Applied

Q71: Richard Feynman emphasized manipulation of matter at the level of _____.

Answer: Atoms

Q72: The concept of building structures atom-by-atom is associated with the _____ approach.

Answer: Bottom-up

Q73: Semiconductor manufacturing commonly follows the _____ approach.

Answer: Top-down

Q74: Bottom-up fabrication resembles natural biological _____.

Answer: Systems

Q75: The exceptional properties of nanomaterials mainly arise because of their extremely small _____.

Answer: Size

Q76: At the nanoscale, physical properties may differ because _____ effects become dominant.

Answer: Quantum

Q77: Nanotechnology enables production of devices requiring less _____.

Answer: Material

Q78: Reduced energy consumption is considered an important _____ of nanotechnology.

Answer: Advantage

Q79: Improved implant surfaces produced through nanotechnology reduce the risk of _____.

Answer: Infection

Q80: Nanotechnology-based drug delivery aims to improve drug _____.

Answer: Absorption

Q81: Nanoparticles used in medicine can improve targeted drug _____.

Answer: Delivery

Q82: Carbon nanotubes possess high strength despite their low _____.

Answer: Weight

Q83: Nano-sized filters remove microorganisms because of their extremely small _____.

Answer: Pores

Q84: Sunscreens containing nanoparticles mainly protect against _____ radiation.

Answer: Ultraviolet

Q85: Nanoclays improve packaging materials by reducing gas _____.

Answer: Permeability

Q86: Nanosensors are capable of detecting extremely _____ concentrations.

Answer: Low

Q87: Chemical nanosensors increase sensitivity because of their high surface _____.

Answer: Area

Q88: One billionth of a meter is defined as one _____.

Answer: Nanometer

Q89: Nanotechnology is considered an interdisciplinary field because it combines multiple _____.

Answer: Disciplines

Q90: Nanobiotechnology combines nanotechnology with _____.

Answer: Biotechnology

Q91: Controlled manipulation at the nanoscale improves the functional _____ of materials.

Answer: Properties

Q92: Nanomaterials may display novel optical, electrical and _____ properties.

Answer: Mechanical

Q93: Biotechnology applications extend to medicine, agriculture and environmental _____.

Answer: Protection

Q94: The production of recombinant insulin is considered an achievement of _____ biotechnology.

Answer: Modern

Q95: Humulin was the first FDA-approved drug produced through recombinant _____ technology.

Answer: DNA

Q96: Nanotechnology contributes to healthcare by improving disease _____.

Answer: Diagnosis

Q97: Lab-on-a-chip devices integrate multiple laboratory functions on a single _____.

Answer: Chip

Q98: The success of nanotechnology largely depends upon controlling material _____.

Answer: Structure

Q99: Nanotechnology enables fabrication of smaller devices without compromising their _____.

Answer: Performance

Q100: The fundamental goal of nanobiotechnology is to apply nanoscale science for biological and _____ applications.

Answer: Medical

Q101: Nanofiltration is primarily employed for the removal of dissolved salts and _____ contaminants.

Answer: Heavy metal

Q102: Nanofibers used in air purification efficiently trap dangerous _____.

Answer: Bacteria

Q103: Gold nanoparticles are utilized in environmental applications to remove volatile _____ compounds.

Answer: Organic

Q104: MCM-41 is also known as self-assembled monolayers on mesoporous _____.

Answer: Supports

Q105: The pore size of MCM-41 generally falls within the _____ nanometer range.

Answer: 10–100

Q106: Replacing metallic automobile components with nanoparticle-reinforced polymers mainly reduces vehicle _____.

Answer: Weight

Q107: A reduction in vehicle weight contributes directly to lower fuel _____.

Answer: Consumption

Q108: One environmental advantage of nanomaterials is the reduction of atmospheric _____.

Answer: Pollution

Q109: Nanotechnology contributes to wastewater treatment mainly through _____.

Answer: Nanofiltration

Q110: Air purification systems containing nanofibers primarily improve indoor air _____.

Answer: Quality

Q111: Quantum dots are classified as inorganic _____.

Answer: Fluorophores

Q112: The core of many quantum dots is composed of cadmium _____.

Answer: Selenide

Q113: CdSe quantum dots are commonly coated with zinc _____.

Answer: Sulphide

Q114: Quantum dots emit fluorescent light when exposed to _____ energy light.

Answer: Low

Q115: The color emitted by a quantum dot mainly depends upon its _____.

Answer: Size

Q116: A polymer coating on quantum dots facilitates attachment of biological _____.

Answer: Conjugates

Q117: Different-sized quantum dots embedded in microbeads produce distinct spectral _____.

Answer: Barcodes

Q118: Quantum dots are valuable because they provide higher fluorescence than conventional _____.

Answer: Markers

Q119: One important biomedical application of quantum dots is biological _____.

Answer: Labeling

Q120: Quantum dots are especially useful for multiplex biological _____.

Answer: Detection

Q121: Biomimetics involves designing materials by imitating _____ systems.

Answer: Biological

Q122: Diatoms possess silica shells that provide both protection and _____.

Answer: Strength

Q123: Some diatoms secrete viscous mucilage that binds colonies _____.

Answer: Together

Q124: The underwater adhesive properties of molluscs serve as a model for _____ materials.

Answer: Self-healing

Q125: Natural underwater adhesives mainly derive their unique properties from specialized _____.

Answer: Proteins

Q126: Sacrificial bonds allow biological adhesive proteins to undergo reversible _____.

Answer: Stretching

Q127: Self-healing materials are inspired by organisms capable of repeated bond breaking and _____.

Answer: Reformation

Q128: Biomimetic materials attempt to reproduce functions already optimized by _____.

Answer: Nature

Q129: The study of natural adhesives contributes to the development of advanced _____.

Answer: Biomaterials

Q130: Diatoms and molluscs are examples illustrating _____-inspired nanotechnology.

Answer: Nature

Q131: Nanofibrous scaffolds possess a very large specific surface _____.

Answer: Area

Q132: Nanofibers are characterized by extremely high _____.

Answer: Porosity

Q133: The pore size of nanofibrous materials is generally _____.

Answer: Small

Q134: The typical diameter of nanofibers ranges from _____ to _____ nm.

Answer: 50–1000

Q135: Annealing nanofibrous structures improves their mechanical _____.

Answer: Stability

Q136: Annealing joins nanofibers mainly at their _____ points.

Answer: Crossing

Q137: Electrospinning is commonly used to fabricate _____.

Answer: Nanofibers

Q138: Both polymer solutions and polymer _____ can be electrospun into nanofibers.

Answer: Melts

Q139: More than _____ different polymers have been successfully electrospun.

Answer: 30

Q140: Polyethylene oxide is one example of a polymer used for _____.

Answer: Electrospinning

Q141: DNA itself can serve as a material for producing _____.

Answer: Nanofibers

Q142: Biological polymers such as collagen can be converted into _____.

Answer: Nanofibers

Q143: Nylon is classified as an _____ polymer used for nanofiber production.

Answer: Organic

Q144: Polyester-based nanofibers belong to the category of _____ polymers.

Answer: Synthetic

Q145: Protein-based nanofibers are especially valuable in _____ applications.

Answer: Biomedical

Q146: High porosity makes nanofibers particularly suitable for tissue _____.

Answer: Engineering

Q147: Nanofibrous scaffolds facilitate cell attachment because of their large surface _____.

Answer: Area

Q148: One important industrial application of nanofibers is advanced _____ systems.

Answer: Filtration

Q149: Electro spun nanofibers combine nanoscale diameter with high _____ strength after stabilization.

Answer: Mechanical

Q150: The unique combination of high surface area, porosity, and nanoscale diameter makes nanofibers highly suitable as biomedical _____.

Answer: Scaffolds

Q151: Nanotechnology in healthcare aims to prevent disease and improve early _____.

Answer: Diagnosis

Q152: Miniaturized probes are primarily developed for recognizing _____.

Answer: Disease

Q153: Nano-dots are mainly used to _____ disease.

Answer: Trace

Q154: MRI in nanomedicine benefits from improved _____ agents.

Answer: Contrast

Q155: CAT stands for Computerized Axial _____.

Answer: Tomography

Q156: Cancer nanomedicines are designed to target only _____ cells.

Answer: Cancer

Q157: Nanotechnology-based antimicrobials are intended to eliminate _____.

Answer: Germs

Q158: Nanotechnology improves orthopedic implants by increasing their _____.

Answer: Compatibility

Q159: Molecules smaller than 100 nm can readily pass through _____ membranes.

Answer: Capillary

Q160: Lipophilic nanoparticles may cross the _____ Barrier.

Answer: Blood-Brain

Q161: Nanoparticles have easier access into _____ than larger particles.

Answer: Cells

Q162: High aspect ratio nanorods penetrate cell membranes more _____.

Answer: Easily

Q163: One important application of nanoparticles inside cells is _____.

Answer: Chemotherapy

Q164: Structural properties of nanomaterials contribute to increased mechanical _____.

Answer: Strength

Q165: Nanomaterials generally exhibit a low coefficient of _____.

Answer: Friction

Q166: Nanowires are used in superconducting magnets for _____ systems.

Answer: MRI

Q167: Nanocomposites improve the performance of X-ray tube _____.

Answer: Anodes

Q168: Carbon nanotubes can function as field emitters for X-ray tube _____.

Answer: Cathodes

Q169: Nanolayers reduce friction in X-ray tube _____.

Answer: Bearings

Q170: Nanoneedles are primarily used as _____.

Answer: Biosensors

Q171: Block copolymers have important applications in tissue _____.

Answer: Engineering

Q172: PET is the abbreviation of Positron Emission _____.

Answer: Tomography

Q173: PAT stands for Photo-Acoustic _____.

Answer: Tomography

Q174: Nanoparticle contrast agents can reduce the need for invasive _____.

Answer: Biopsies

Q175: Ultrasound nanoparticle contrast agents improve detection of vascular _____.

Answer: Thrombi

Q176: Nanowires containing receptors for PSA are useful in detecting _____ cancer.

Answer: Prostate

Q177: PSA stands for Prostate Specific _____.

Answer: Antigen

Q178: Nanowire-based biosensors can detect markers from as few as _____ tumor cells.

Answer: Ten

Q179: Carbon nanotubes can transport _____ into cells.

Answer: Proteins

Q180: Quantum dots are semiconductor nanoparticles that emit different _____.

Answer: Colors

Q181: The emitted color of a quantum dot mainly depends on its _____.

Answer: Size

Q182: Compared with conventional dyes, quantum dots exhibit a much longer _____.

Answer: Lifetime

Q183: One advantage of quantum dots is the availability of _____ fluorescent colors.

Answer: More

Q184: Quantum dots are widely used as biological _____ devices.

Answer: Tracking

Q185: Nanotechnology enables medicines to be delivered to the _____ location.

Answer: Exact

Q186: Targeted nanodrug delivery minimizes unwanted _____ effects.

Answer: Side

Q187: Organic dendrimers are artificial molecules approximately the size of a _____.

Answer: Protein

Q188: Gold-coated glass beads used in drug delivery are sensitive to _____ infrared light.

Answer: Near

Q189: Nanotechnology can be used for the destruction of harmful _____ organisms.

Answer: Eukaryotic

Q190: Artificial red blood cells are proposed for efficient transport of _____.

Answer: Oxygen

Q191: Nanorobot-containing creams may remove excess skin _____.

Answer: Oils

Q192: Nanorobots are proposed to restore missing natural _____ compounds in skin.

Answer: Moisturizing

Q193: Deep pore cleaning is considered possible because nanorobots can physically enter _____.

Answer: Pores

Q194: Dental nanorobots are designed to clean surfaces near the _____ line.

Answer: Gum

Q195: Medical nanodevices may strengthen the body's _____ system.

Answer: Immune

Q196: Bloodstream nanodevices may remove atherosclerotic _____.

Answer: Deposits

Q197: Future nanoprobes are predicted to undergo self-_____ similar to mitosis.

Answer: Replication

Q198: Nanodiagnostics mainly applies nanotechnology principles for medical _____.

Answer: Diagnosis

Q199: Gold nanoparticles are among the most frequently used nanoparticles in _____.

Answer: Diagnostics

Q200: According to the BT405 handouts, nanodiagnostics involves the manipulation and assessment of single _____.

Answer: Molecules

Q201: Nanomedicine combines nanotechnology with _____ science.

Answer: Medical

Q202: The major objective of targeted drug delivery is to increase therapeutic _____.

Answer: Efficiency

Q203: Nanocarriers reduce drug toxicity by limiting exposure to healthy _____.

Answer: Tissues

Q204: Nanoparticles used in cancer therapy preferentially accumulate in _____ tissues.

Answer: Tumor

Q205: Controlled drug release helps maintain an effective drug _____.

Answer: Concentration

Q206: Polymeric nanoparticles are widely used because of their excellent _____ compatibility.

Answer: Biocompatibility

Q207: Biodegradable nanocarriers gradually decompose into _____ products.

Answer: Non-toxic

Q208: Surface modification of nanoparticles improves their biological _____.

Answer: Performance

Q209: Ligands attached to nanoparticle surfaces improve target cell _____.

Answer: Recognition

Q210: Nanoparticles entering cells mainly occur through _____.

Answer: Endocytosis

Q211: Endocytosis allows nanoparticles to enter the cellular _____.

Answer: Cytoplasm

Q212: Nanoparticle size greatly influences cellular _____.

Answer: Uptake

Q213: Smaller nanoparticles generally possess a larger surface-area-to-_____ ratio.

Answer: Volume

Q214: Increased surface area enhances nanoparticle _____.

Answer: Reactivity

Q215: One limitation of nanoparticles is their possible long-term _____.

Answer: Toxicity

Q216: Nanotoxicology studies the harmful effects of _____ materials.

Answer: Nanostructured

Q217: Reactive oxygen species generated by nanoparticles may cause oxidative _____.

Answer: Stress

Q218: Oxidative stress primarily damages cellular _____.

Answer: Components

Q219: Nanoparticle toxicity depends upon size, shape and surface _____.

Answer: Chemistry

Q220: Biocompatibility is an essential property for nanoparticles used in _____.

Answer: Medicine

Q221: Nanobiosensors convert biological responses into measurable _____.

Answer: Signals

Q222: The recognition element of a biosensor specifically binds the target _____.

Answer: Analyte

Q223: Enzyme-based biosensors depend upon catalytic _____.

Answer: Activity

Q224: Antibody-based biosensors utilize antigen-antibody _____.

Answer: Interaction

Q225: DNA biosensors detect complementary nucleic acid _____.

Answer: Sequences

Q226: The transducer converts a biological event into an electrical _____.

Answer: Signal

Q227: Nanobiosensors are valued because of their high _____.

Answer: Sensitivity

Q228: One major application of biosensors is rapid disease _____.

Answer: Diagnosis

Q229: Carbon nanotubes improve biosensor performance due to their excellent electrical _____.

Answer: Conductivity

Q230: Gold nanoparticles improve biosensor sensitivity by enhancing signal _____.

Answer: Detection

Q231: Nanobiotechnology integrates nanotechnology with modern _____.

Answer: Biotechnology

Q232: Biofunctionalization involves attaching biological molecules onto nanoparticle _____.

Answer: Surfaces

Q233: Surface functionalization increases nanoparticle _____.

Answer: Specificity

Q234: Nanomaterials intended for medical use should exhibit minimal _____ response.

Answer: Immune

Q235: Drug-loaded nanoparticles are designed to improve patient _____.

Answer: Compliance

Q236: Encapsulation protects therapeutic agents from premature _____.

Answer: Degradation

Q237: Sustained drug release decreases the frequency of drug _____.

Answer: Administration

Q238: Nanotechnology enhances pharmaceutical formulations by improving drug _____.

Answer: Stability

Q239: Poorly soluble drugs often show improved bioavailability after _____ formulation.

Answer: Nano

Q240: One important advantage of nanoformulations is increased therapeutic _____.

Answer: Effectiveness

Q241: Tissue engineering combines scaffolds, cells and biologically active _____.

Answer: Molecules

Q242: Nanofibrous scaffolds imitate the natural extracellular _____.

Answer: Matrix

Q243: The extracellular matrix provides structural support for _____.

Answer: Cells

Q244: Scaffold porosity directly influences cell _____.

Answer: Growth

Q245: Biodegradable scaffolds gradually disappear as new tissue is _____.

Answer: Formed

Q246: An ideal scaffold should promote cell attachment and _____.

Answer: Proliferation

Q247: Nanotechnology contributes significantly to regenerative _____.

Answer: Medicine

Q248: Stem-cell growth can be influenced by nanoscale surface _____.

Answer: Topography

Q249: Surface roughness at the nanoscale affects cellular _____.

Answer: Adhesion

Q250: The primary goal of nanobiotechnology in regenerative medicine is functional tissue _____.

Answer: Repair

Q251: The extracellular matrix (ECM) mainly provides structural _____ for surrounding cells.

Answer: Support

Q252: Nanostructured scaffolds improve tissue regeneration by mimicking the natural cellular _____.

Answer: Environment

Q253: Cell differentiation can be influenced by nanoscale surface _____.

Answer: Features

Q254: Biocompatible scaffolds should promote cell attachment without causing _____.

Answer: Toxicity

Q255: One objective of scaffold engineering is to support nutrient _____.

Answer: Diffusion

Q256: Nanostructured biomaterials accelerate wound _____.

Answer: Healing

Q257: An ideal scaffold gradually degrades as new tissue is _____.

Answer: Formed

Q258: Controlled degradation prevents premature scaffold _____.

Answer: Failure

Q259: Surface modification improves interaction between cells and _____.

Answer: Biomaterials

Q260: Nanofibrous scaffolds provide a high surface area for cellular _____.

Answer: Attachment

Q261: One challenge in nanotechnology is large-scale _____.

Answer: Manufacturing

Q262: Commercial production of nanomaterials requires process _____.

Answer: Standardization

Q263: Reproducibility is essential for successful nanomaterial _____.

Answer: Fabrication

Q264: Quality control is necessary to maintain nanoparticle _____.

Answer: Uniformity

Q265: Regulatory approval of nanomedicines requires extensive _____ evaluation.

Answer: Safety

Q266: Toxicological assessment is necessary before clinical _____.

Answer: Application

Q267: Environmental concerns regarding nanotechnology mainly involve nanoparticle _____.

Answer: Disposal

Q268: Sustainable nanotechnology focuses on minimizing environmental _____.

Answer: Impact

Q269: Risk assessment evaluates both hazards and potential _____.

Answer: Exposure

Q270: Safe handling procedures reduce occupational nanoparticle _____.

Answer: Exposure

Q271: Nanotechnology has applications in medicine, agriculture, electronics and _____ science.

Answer: Environmental

Q272: Smart nanomaterials respond to changes in their _____.

Answer: Environment

Q273: Stimuli-responsive nanomaterials may react to changes in pH, temperature or _____.

Answer: Light

Q274: Multifunctional nanoparticles can simultaneously perform diagnosis and _____.

Answer: Therapy

Q275: The combination of diagnosis and therapy is known as _____.

Answer: Theranostics

Q276: Theranostic nanoparticles improve personalized _____.

Answer: Medicine

Q277: Nanotechnology enables earlier detection of disease using highly sensitive _____.

Answer: Biosensors

Q278: Molecular imaging provides visualization at the _____ level.

Answer: Cellular

Q279: Target-specific nanoparticles reduce damage to _____ tissues.

Answer: Healthy

Q280: Controlled release formulations help maintain consistent drug _____.

Answer: Levels

Q281: Surface-functionalized nanoparticles selectively bind specific cellular _____.

Answer: Receptors

Q282: Ligand-receptor interaction improves nanoparticle _____ specificity.

Answer: Targeting

Q283: Nanocarriers improve therapeutic efficiency by enhancing drug _____.

Answer: Bioavailability

Q284: Polymeric nanocarriers protect drugs from enzymatic _____.

Answer: Degradation

Q285: Liposomal systems are examples of _____ delivery systems.

Answer: Drug

Q286: Biodegradable nanoparticles reduce long-term tissue _____.

Answer: Accumulation

Q287: One future application of nanotechnology is personalized _____.

Answer: Medicine

Q288: Nanotechnology may improve vaccine _____ systems.

Answer: Delivery

Q289: Nanoparticle-based vaccines enhance antigen _____.

Answer: Presentation

Q290: Diagnostic nanoparticles increase detection _____ for disease biomarkers.

Answer: Sensitivity

Q291: Nanomaterials are increasingly used to improve medical _____.

Answer: Imaging

Q292: Functional nanomaterials combine structural and biological _____.

Answer: Properties

Q293: One advantage of nanoscale materials is their tunable physical and chemical _____.

Answer: Properties

Q294: Future nanotechnology emphasizes safe, effective and _____ applications.

Answer: Sustainable

Q295: Ethical concerns regarding nanotechnology include human health and environmental _____.

Answer: Safety

Q296: Responsible development of nanotechnology requires scientific and regulatory _____.

Answer: Oversight

Q297: Translation of nanotechnology from laboratory research to industry requires technology _____.

Answer: Transfer

Q298: Successful nanotechnology commercialization depends upon product _____.

Answer: Reliability

Q299: Continued advancement in nanobiotechnology depends on interdisciplinary _____.

Answer: Research

Q300: The ultimate objective of nanobiotechnology is to improve human life through innovative nanoscale _____.

Answer: Technologies

🚨 Concept Cracker MCQS Most Important Repeated Conceptual 🚨

⚠ Important Note:

These MCQs have been compiled from repeated exam patterns, important concepts, and previous assessments to support your preparation.

📚 However, do NOT rely solely on this compilation.

For the best results, thoroughly study your Handouts, PPTs, Video Lectures, and Course Material as well.

Prepare Smart, Work Hard, and Aim for Excellence.

Best of Luck! 🍀📚

BS Biological Sciences Team 🚀🎓

JAWAD CHAUDHARY